ACTIVITY 3 B

ORGANIC CHEMISTRY

BOLD DESIGN STUDIO

2-methylpropane

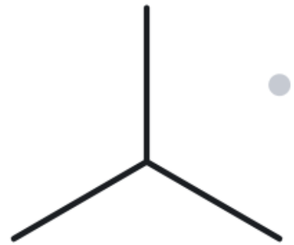
Name decoded

Parent chain: 3 Carbon

Substituent: Methyl on C-2

Expected Formula: C_4H_{10}

Bond line structure



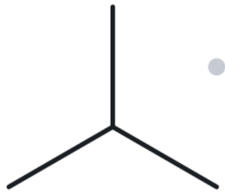
Formula Check

Counted from the drawing

C-4

H-10

Back check



2-methylpropane

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

3-ethylpentane

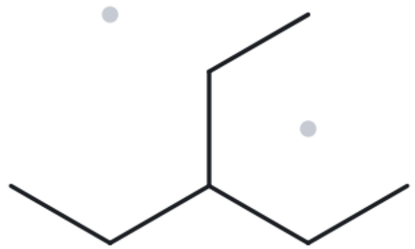
Name decoded

Parent chain: 5 Carbon

Substituent: Ethyl on C-3

Expected Formula: C_7H_{16}

Bond line structure



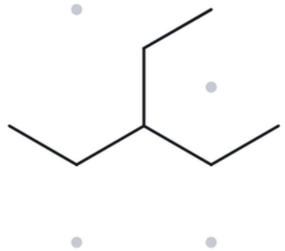
Formula Check

Counted from the drawing

C-7

H-16

Back check



✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

2,2-dimethylbutane

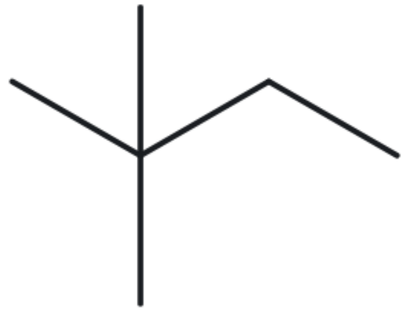
Name decoded

Parent chain: 4 Carbon

Substituent: 2 methyl on C-2

Expected Formula: C_6H_{14}

Bond line structure



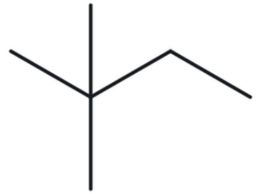
Formula Check

Counted from the drawing

C-6

H-14

Back check



2,2-dimethylbutane

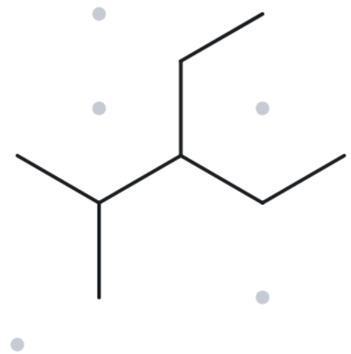
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

3-ethyl-2-methylpentane

Name decoded

Parent chain: 5 Carbon
Substituent: ethyl on C-3, methyl on C-2
Expected Formula: C_8H_{18}

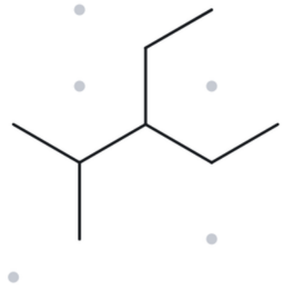
Bond line structure



Formula Check

Counted from the drawing
C-8
H-18

Back check



3-ethyl-2-methylpentane

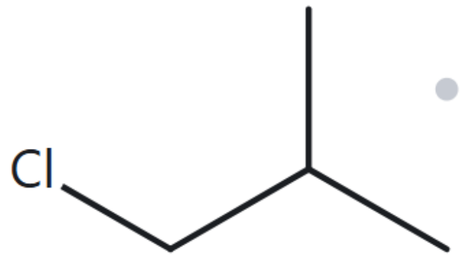
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

1-chloro-2-methylpropane

Name decoded

Parent chain: 3 Carbon
Substituent: Chloro on C-1, methyl on C-2
Expected Formula: C_4H_9Cl

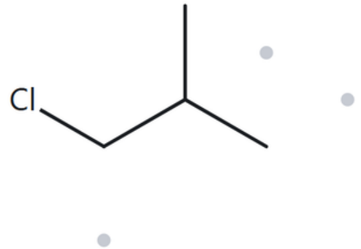
Bond line structure



Formula Check

Counted from the drawing
C-4
H-9
Cl-1

Back check



✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

1,2-dimethylcyclopentane

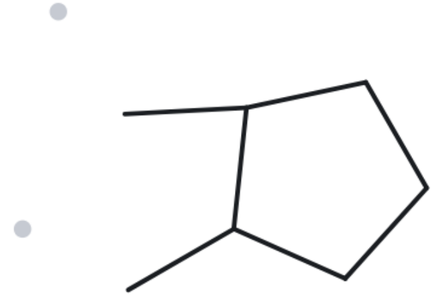
Name decoded

Parent chain: 5 Carbon ring

Substituent: methyl on C-1, and C-2

Expected Formula: C_7H_{14}

Bond line structure



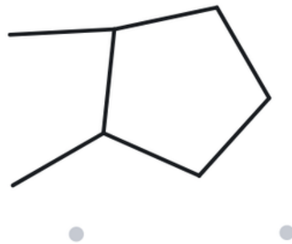
Formula Check

Counted from the drawing

C-7

H-14

Back check



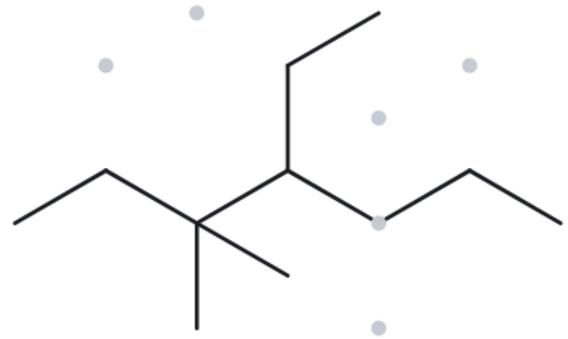
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

4-ethyl-3,3-dimethylheptane

Name decoded

Parent chain: 7 Carbon
Substituent: ethyl on C-4, and 2 methyl on C-3
Expected Formula: $C_{11}H_{24}$

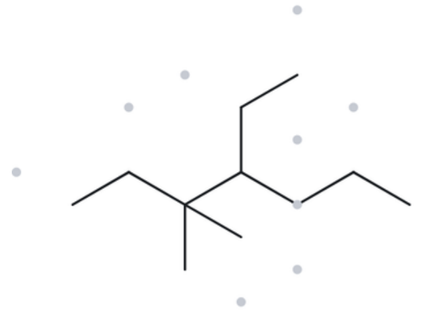
Bond line structure



Formula Check

Counted from the drawing
C-11
H-24

Back check



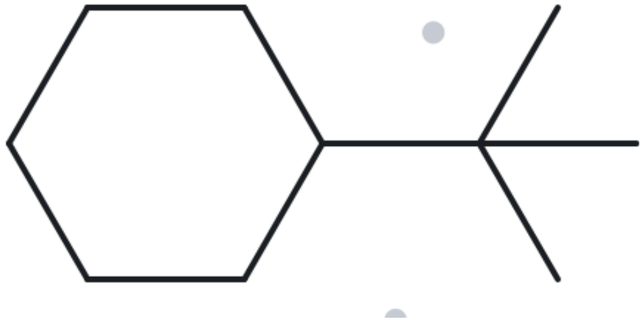
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

tert-butylcyclohexane

Name decoded

Parent chain: 6 Carbon ring
Substituent: tert-butyl group
Expected Formula: $C_{10}H_{20}$

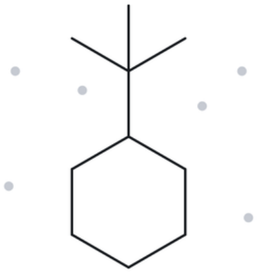
Bond line structure



Formula Check

Counted from the drawing
C-10
H-20

Back check



(1,1-dimethylethyl)cyclohexane

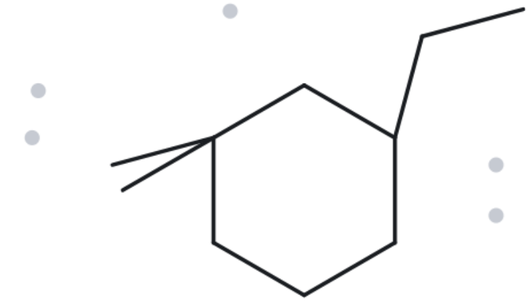
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

3-ethyl-1,1- dimethylcyclohexane

Name decoded

Parent chain: 6 Carbon ring
Substituent: ethyl on C-3, 2 methyl on C-1
Expected Formula: $C_{10}H_{20}$

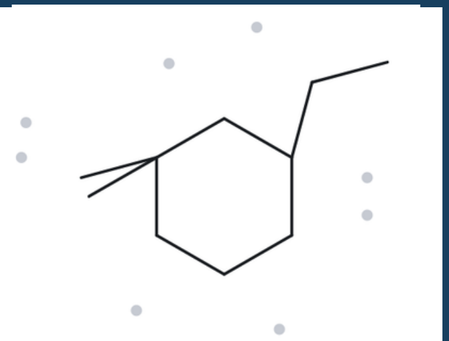
Bond line structure



Formula Check

Counted from the drawing
C-10
H-20

Back check



✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

2-ethylbutane(3-methylpentane)

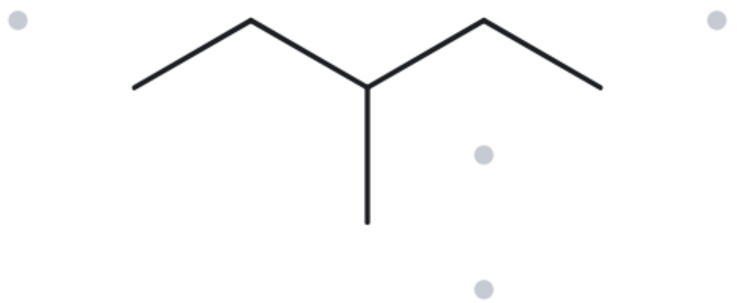
Name decoded

Parent chain: 5 Carbon

Substituent: methyl on C-3

Expected Formula: C_6H_{14}

Bond line structure



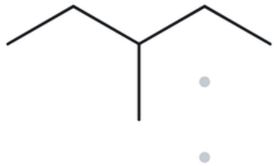
Formula Check

Counted from the drawing

C-6

H-14

Back check



3-methylpentane

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded